Few-Shot Oriented Object Detection in Remote Sensing Images via Memorable Contrastive Learning

Jiawei Zhou, Wuzhou Li, Yi Cao, Hongtao Cai, Tianjin Huang, Gui-Song Xia, *Senior Member, IEEE*, and Xiang Li, *Member, IEEE*

***Abstract*— Few-shot object detection (FSOD) has attracted significant research attention in remote sensing due to its potential to reduce reliance on large annotated datasets. However, two challenges remain in this area: 1) axis-aligned proposals, which can result in misalignment for arbitrarily oriented objects and 2) object misclassification due to limited annotated data, which hinders generalization to unseen classes. To address these issues, we propose a novel method for few-shot oriented object detection (FSOOD) in remote sensing images. Our approach employs oriented bounding boxes (OBBs) instead of horizontal ones to learn more effective feature representations for arbitrarily oriented aerial objects, enhancing detection accuracy. Additionally, we introduce a supervised contrastive learning module with a dynamically updated memory bank, enabling the model to leverage large batches of negative samples and to better learn discriminative features for unseen classes. Extensive experiments on DOTA, HRSC2016, and DIOR-R datasets demonstrate superior performance of our proposed method in FSOOD. Code and pretrained models are publicly available at https://github.com/BriFuture/fomc.**

***Index Terms*— Contrastive learning, few-shot object detection (FSOD), oriented object detection, remote sensing images.**

I. Introduction

OBJECT detection is a long-standing task in remote sensing, which has wide real-world applications such as urban planning, environmental management, and traffic monitoring [1], [2], [3], [4], [5]. Deep learning approaches have made significant progress in object detection [6]. Despite their success, they often require substantial amounts of labeled data to perform well [7], [8], which are costly and labor-intensive to obtain [9]. Few-shot object detection (FSOD) has emerged as a promising direction to mitigate the dependency on large-scale datasets, aiming at detecting unseen objects using only a few examples [10], [11], [12], [13]. However, traditional FSOD methods still face two challenges in the remote sensing field:

1) *Arbitrarily Oriented Objects:* Different from natural scene images, remote sensing images are typically captured from a bird's-eye view, causing objects to appear in arbitrary orientations. Existing FSOD methods are primarily designed for horizontal bounding box (HBB) detection, which may inadvertently cover adjacent instances. As shown in Fig. 1, HBBs can cause ambiguity when classifying densely packed objects, especially in limited-data scenarios.
2) *Misclassification of Novel Objects:* In few-shot detection, models can often localize the novel objects correctly but misclassify them into confusing classes. Misclassification constitutes a major source of errors in FSOD [14]. Learning a well-separated decision boundary for novel objects remains a challenging issue due to the limited training data.

In response to these challenges, we introduce a novel approach named few-shot oriented object detection with memorable contrastive learning (FOMC). First, we adopt oriented bounding boxes (OBBs) instead of HBBs to tackle the FSOD

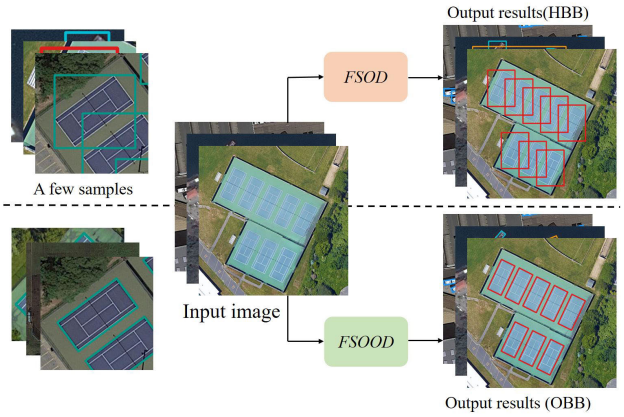


Fig. 1. Comparison between FSOD using HBB and FSOOD using OBB. HBBs often include background or neighboring objects, while OBBs offer tighter, more precise localization for oriented objects.

Received 13 November 2024; revised 30 March 2025 and 1 June 2025; accepted 4 June 2025. Date of publication 11 June 2025; date of current version 15 July 2025. This work was supported by the Special Fund of Hubei Luojia Laboratory under Grant 220100013. *(Corresponding author: Xiang Li.)*

Jiawei Zhou and Hongtao Cai are with the School of Electronic Information, Wuhan University, Wuhan 430072, China, and also with Hubei Luojia Laboratory, Wuhan 430072, China.

Wuzhou Li is with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430072, China.

Yi Cao is with the School of Electronic Information, Wuhan University, Wuhan 430072, China.

Tianjin Huang is with the Department of Computer Science, University of Exeter, EX4 4QF Exeter, U.K.

Gui-Song Xia is with the School of Computer Science and the State Key Laboratory of Information Engineering, Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430072, China.

Xiang Li is with the Department of Computer Science, University of Reading, RG6 6AH Reading, U.K. (e-mail: xiangli92@ieee.org).

This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2025.3578632, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2025.3578632

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

problem in remote sensing images. OBBs provide tighter anchor alignment for oriented objects and can potentially improve detection performance by reducing the interference of complex backgrounds. Second, we introduce a memorable contrastive learning (MCL) module to facilitate the learning of distinctive feature representations. Contrastive learning aims to map positive (similar) pairs closely in the embedding space, while positioning negative (dissimilar) pairs further apart. However, in the FSOD task, the number of positive samples for novel objects is limited. We use the memory bank to store samples, enhancing contrastive learning. Furthermore, we observe that unselected instances in the few-shot setting can lead to significant performance degradation. To address this issue, we introduce a simple and effective instance masking module to eliminate the confusion caused by unselected instances.

Our main contributions can be summarized as follows.

1) We introduce FOMC, a few-shot oriented object detector for remote sensing images. Instead of traditional HBB detection, our model effectively detects oriented objects in few-shot scenarios.
2) We propose the MCL module, a contrastive learning-based approach with a dynamically updated memory bank. This mechanism guides the model to learn more discriminative features with larger intraclass compactness and interclass differences by covering a richer set of negative samples across different mini-batches.
3) We develop a simple yet effective instance masking module to eliminate the confusion caused by unselected instances in few-shot settings, significantly enhancing our model's performance.
4) Experiments on DOTA [8], HRSC2016 [15], and DIOR-R [16] datasets demonstrate superior performance in few-shot oriented object detection (FSOOD) in remote sensing images.

II. Related Works

A. Oriented Object Detection in Remote Sensing Images

With the significant advancements in deep learning, numerous methods have been developed to perform object detection in natural images. Existing deep learning-based methods can be broadly classified into two-stage and one-stage detectors. Two-stage detectors generate region proposals that potentially contain objects and then refine the location and make the final prediction based on region proposals. Representative two-stage detectors include R-CNN [6], SPPNet [17], Fast R-CNN [18], and Faster R-CNN [19]. In contrast, one-stage detectors complete the object detection prediction in one step. YOLO [20], SSD [21], RetinaNet [22], CornerNet [23], CenterNet [24], and transformer-based DETR [25] are representative one-stage detectors.

Unlike natural scene images, aerial images generally present unique challenges in the object detection task. Remote sensing images are captured from a bird's-eye view, and ground objects typically exhibit arbitrary orientations. Wu et al. [26] embed a smaller U-Net within a larger U-Net backbone, enabling multilevel and multiscale representation learning for infrared small object detection. Li et al. [27] propose a lightweight recursive refinement network that leverages recursive feature fusion and attention mechanisms to achieve high-precision detection of small objects in remote sensing images. However, the widely used HBBs in general object detection models fail to locate rotated objects accurately. Therefore, oriented object detection has garnered significant research attention in remote sensing [28], [29]. These methods are typically categorized into rotation-aware and rotation-invariant feature learning. Notably, ROI-Transformer [30] introduces a rotation-aware RoI Align to learn rotation-invariant features. Yang et al. [31] introduce SCRDet to detect small, cluttered objects in arbitrary orientations. Xu et al. [32] propose to glide the vertex of the HBB along their sides to locate an oriented object accurately. To mitigate the performance degradation caused by the discontinuity of loss in oriented object detection, Qian et al. [33] design a modulated rotation loss and propose an eight-parameter detector to directly predict the corners of the objects. Other studies focus on learning rotation-invariant features that remain consistent under various rotational transformations. Yao et al. [34] utilize class-aware points to encode the rotation-insensitive representation of arbitrary-oriented objects. Han et al. [35] propose ReDet, which extracts both rotation-equivariant and rotation-invariant features according to the orientation of RoI to accurately predict the oriented objects. Oriented R-CNN [36] introduces an oriented region proposal network (RPN) and an oriented R-CNN head to enhance performance on highly rotated objects. Qian et al. [37] propose a unified transferring strategy that enables the efficient adaptation of bounding box regression losses from horizontal to oriented object detection, leading to superior OBB detection performance. Yao et al. [38] propose a centric probability-based sample selection framework that improves OBB detection by assigning spatially informed confidence scores, dynamically optimizing sample boundaries, and enhancing sample diversity through probabilistic sampling.

B. Few-Shot Object Detection in Remote Sensing Images

Current FSOD methods are typically divided into two categories: meta-learning-based [10], [39] methods and transfer learning-based [12], [40], [41] methods. Meta-learners aim to "learn how to learn" by designing and solving a series of FSOD tasks, enabling a model to adapt and generalize to novel data more efficiently. This process typically involves a complex data organization and training pipeline, making it challenging to apply in practical scenarios. On the other hand, transfer-learning-based methods aim to leverage knowledge learned from abundant base-class data and fine-tune the model with a few samples from novel classes. For example, TFA [40] adopts a two-stage "pretrain–fine-tune" paradigm: pretrain the model on sufficient base-class data and then fine-tune the last few layers using novel class data. Despite its simplicity, TFA outperforms meta-learning approaches [40]. In this study, we adopt the two-stage fine-tuning strategy to achieve simple and effective FSOD in remote sensing images.

FSODM [10] is a pioneering work in FSOD in remote sensing, which develops a multiscale object detector with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

a feature reweighting module that assigns varying weights to support image features. Wolf et al. [11] introduce a double-head predictor to decouple the prediction for base classes and novel classes, preventing performance degradation. Cheng et al. [12] propose a prototype-CNN to convert support images to class-specific prototypes, and fuse the information to generate better proposals, further enhancing detection performance. Xiao et al. [42] present a self-adaptive attention network to fully leverage the instance-level relations for better detection results. Zhou et al. [13] develop an FSOD model for scale-variant remote sensing images by incorporating context-aware feature aggregation. Zhao et al. [43] utilize the involution operator in the backbone network to enhance object classification and design a feature aggregation module for multiscale detection. Li et al. [44] leverage metric learning and develop a multisimilarity module to learn distinctive feature representations, achieving impressive FSOD performance on aerial imagery. Zhang et al. [45] address the issue of catastrophic forgetting by proposing a representation compensation module to decouple the prediction of base and novel classes. Lu et al. [46] propose an FSOD method by introducing text-modal knowledge and designing a cross-modal fusion module to integrate text and visual features. Zhang et al. [47] incorporate a self-training-based approach into the fine-tuning procedure to address the confusion caused by unlabeled objects in FSOD. To eliminate the interference caused by noise attributes and enhance FSOD performance, Wu et al. [48] extract refined features as the representatives of each category instead of directly using coarse features. However, these methods generally overlook the detection of arbitrarily oriented objects. Recent work by Su et al. [49] makes the first attempt to address oriented object detection in a few-shot setting using a multioriented enhancement module to handle scale variations. Our research aims to tackle the challenges of object misclassification in the FSOOD task.

C. Contrastive Learning

The main idea of contrastive learning is to learn discriminative representations by minimizing the distances between positive pairs while simultaneously maximizing distances between negative ones in the embedding space. Prior self-supervised contrastive learning methods take different augmented views of the same data as positive samples, and randomly picked objects from the mini-batch as negative samples. For example, SimCLR [50] introduces a straightforward framework that computes pairwise similarities among images within the mini-batch. On the other hand, MoCo [51] devises a momentum-updated dictionary with a queue, enabling the utilization of considerably large batches of negative samples. The increasing number of negative samples improves the discrimination ability between positive and negative samples, as demonstrated in several papers [51], [52], [53].

Supervised contrastive learning methods leverage label information to achieve superior performance. These methods regard positive samples as those with the same label, and samples from different classes naturally serve as negatives. However, most prior research focuses on image classification [53] and identification [54], and merely a few works adopt it for the FSOD task. FSCE [14] proposes a contrastive proposal encoding branch to learn discriminative feature embeddings and enhance the detection performance for natural scene images. Huang et al. [55] propose an FSOD model with dual-contrastive learning, which consists of a supervised contrastive branch and a prototypical contrastive learning module to learn good representations to boost the performance for classification and then reserve the learned information to maintain the performance for the base classes. In the few-shot setting, positive samples are scarce for novel instances, making it crucial to fully utilize negative samples for learning discriminative features. Inspired by MoCo, our approach adopts a memory bank to enable sufficient contrastive learning in the FSOD task.

III. Methodology

We present our proposed FOMC in this section. Section III-A describes the problem setting for the FSOOD task. Section III-B provides an overview of our proposed method. The MCL module and the instance masking module are discussed in Sections III-C and III-D, respectively. Finally, the training strategy is described in Section III-E.

A. Few-Shot Oriented Object Detection

Our goal is to obtain an oriented object detector for aerial objects with a few novel annotated samples. The overall classes of the dataset are divided into base classes C_{base} and novel classes C_{novel}. Consequently, the dataset can be segmented into D_{base} and D_{novel} subdatasets. D_{base} subdataset contains all data belonging to base classes C_{base}, while D_{novel} subdataset is a small dataset containing equally sampled shots from all classes, including C_{base} and C_{novel}.

B. Method Overview

Our FOMC employs a two-stage training paradigm, as shown in Fig. 2. In the base-training stage, a base detector is trained with abundant base-class data. Note that our method does not require specific architectures for base detectors; it is compatible with commonly used single-stage detectors (e.g., S^2A-Net) or two-stage detectors (e.g., Oriented R-CNN). Following the design of TFA [40], our method decouples classification and localization, allowing for improved feature learning for classification without affecting the performance of the localization task.

In the fine-tuning stage, the base detector is trained with a few annotated samples. Specifically, a small balanced dataset is created by randomly sampling K ($K = 5, 10, 20, 30$) shots of instances from both C_{base} and C_{novel} classes. To enable novel class detection, we modify the box head by adding randomly initialized weights for novel classes and fine-tuning the classification and regression networks. During this stage, the backbone network is frozen to prevent overfitting, while the remaining parts are trained simultaneously to learn object detection for both base and novel classes to achieve overall performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

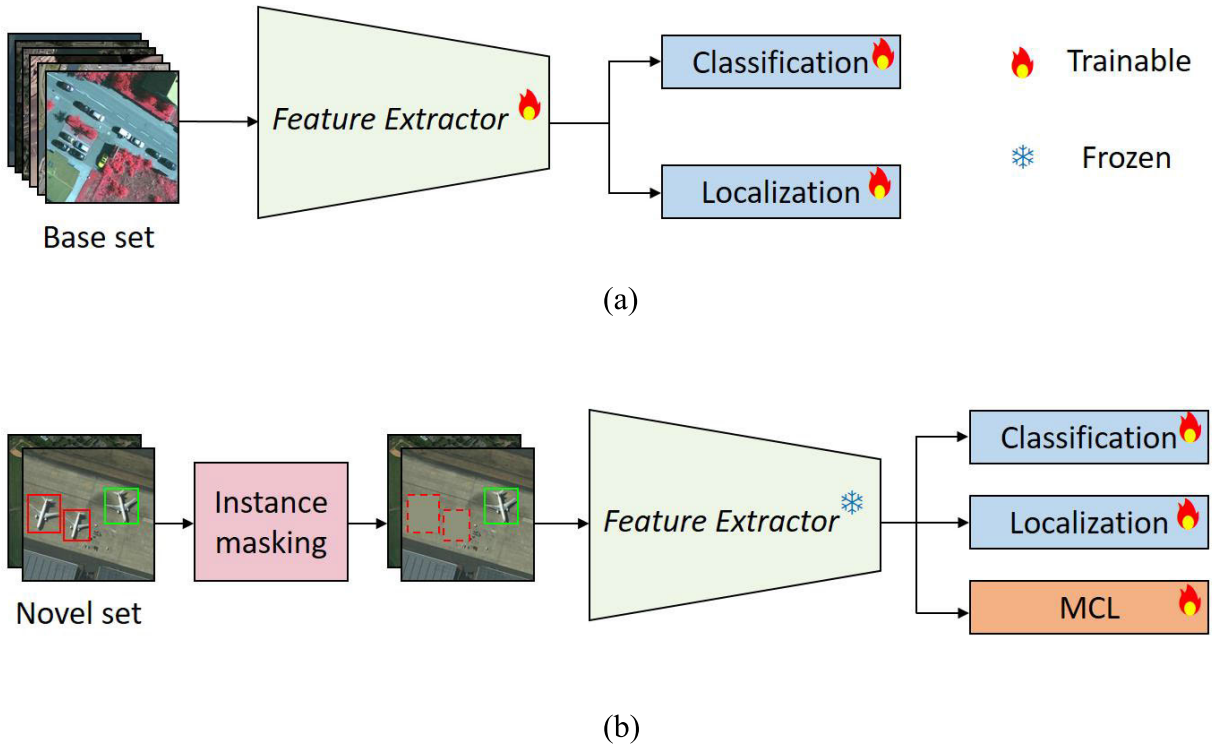


Fig. 2. Overall architecture of our proposed FOMC model. (a) In the base-training phase, all network parameters of the model are trained using abundant data from the base classes. (b) In the fine-tuning stage, the parameters of the ResNet backbone network are frozen, while the other modules are trained at a lower learning rate using a few samples from novel classes. An instance masking module is designed to eliminate confusing labels, and an MCL module is designed to encourage class-distinctive feature learning via contrastive loss.

C. MCL Module

As shown in Fig. 3, our MCL module consists of a projection encoder and a memory bank. The projection encoder comprises two convolutional layers with one ReLU activation layer between them. The encoder maps rotation-invariant proposal features to an embedding space. The memory bank stores proposal features, IoUs, and the corresponding ground truth labels for subsequent contrastive learning. Instead of employing momentum-updated teacher strategies, our MCL module simply applies stepwise weighting to proposals throughout the fine-tuning process.

1) Object Proposal Memory Bank: We adopt a memory bank to store object proposal information, differing from conventional approaches that store only feature embeddings. In addition to the feature embeddings of object proposals, the memory bank stores their IoU scores with ground truth boxes, the corresponding ground truth labels, and the training steps at which they were generated. The implementation of this memory bank significantly expands the range of instances available for contrastive learning, see Fig. 3 for illustration. Consequently, this enhancement enables the model to better distinguish object features, particularly under few-shot learning conditions. To accommodate the expanded dataset, the memory bank is configured with a capacity significantly larger than the mini-batch size.

2) Contrastive Loss: Drawing on the principles of supervised contrastive objectives in classification [53] and identification [54], we formulate an instance-level contrastive loss based on batchwise proposals and those in the memory bank. This loss is specifically designed to promote intraclass similarity while simultaneously enhancing interclass separability, fostering a more robust and discriminative learning model. Specifically, given a batch of M proposals, $\{x_i, c_i, y_i\}_{i=1}^{M}$, where x_i is the orientation-invariant feature embedding for the ith object proposal, c_i denotes the IoU between the proposal and its matched ground truth box, and y_i represents the ground truth label. All proposal features are normalized before calculating the contrastive loss.

Our MCL contrastive loss is formulated as follows:

$$\mathcal{L}_{\mathrm{MCL}} = \frac{1}{M}\sum_{i=1}^{M} \phi(c_i) \cdot \left(\mathcal{L}_i^{\mathrm{in}} + \mathcal{L}_i^{\mathrm{cr}}\right) \tag{1}$$

where $\mathcal{L}_i^{\mathrm{in}}$ and $\mathcal{L}_i^{\mathrm{cr}}$ denote the in-batch and cross-batch contrastive losses for the ith proposal, respectively. $\phi(c_i)$ is a weighting function that controls the consistency of proposals, calculated as follows:

$$\phi(c_i) = \mathbb{I}(c_i > \theta) \cdot c_i \tag{2}$$

where $\mathbb{I}$ denotes the indicator function, which outputs 1 when the condition holds and 0 otherwise, and θ denotes the IoU threshold. By default, we set θ to 0.5 in our experiments. In this way, only the most centered object proposals will be included for model training.

The in-batch contrastive loss is calculated as

$$\mathcal{L}_i^{\mathrm{in}} = \frac{-1}{M}\sum_{j=1}^{M} \log \frac{\exp(x_i \cdot x_j/\tau)}{\sum_{k=1,k\neq i}^{M} \exp(x_i \cdot x_k/\tau)} \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

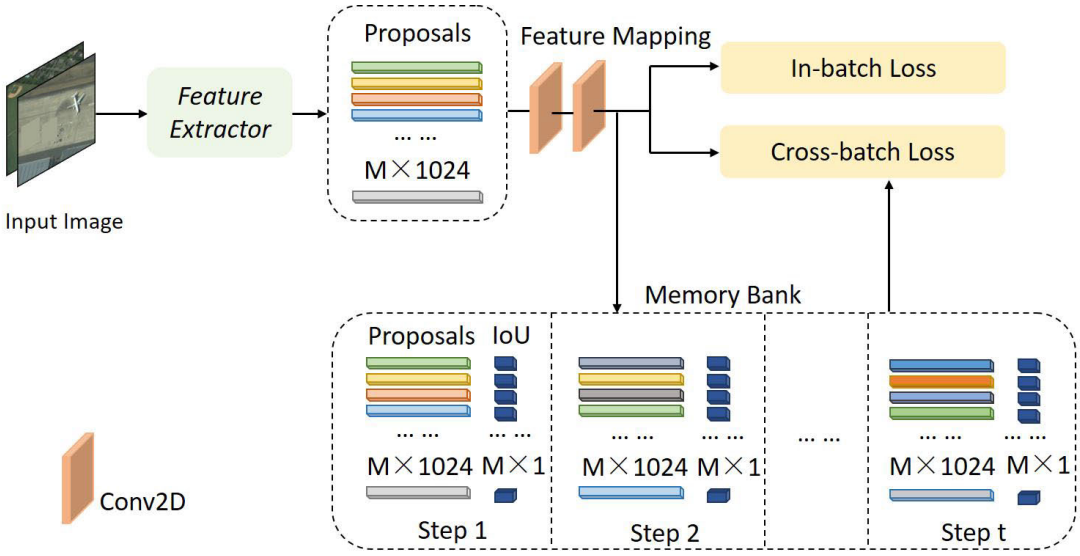


Fig. 3. MCL encoding module. The memory bank stores for each proposal the feature embeddings and the IoU score between the proposal and the matched ground truth box.

and the cross-batch contrastive loss is calculated as

$$\mathcal{L}_i^{\text{cr}} = \frac{-1}{N} \sum_{j=1}^{N} w_{ij} \cdot \log \frac{\exp(x_i \cdot z_j/\tau)}{\sum_{k=1}^{N} \exp(x_i \cdot z_k/\tau)} \tag{4}$$

where τ denotes a scale factor, N denotes the size of the memory bank, and z_j denotes the feature embedding stored in the memory bank. w_{ij} denotes the weight assigned to the jth proposal in the memory bank with respect to the ith proposal in the current batch, formulated as follows:

$$w_{ij} = \max(w_0 - \alpha \cdot (t_i - t_j),\ 0) \tag{5}$$

where t_i denotes the training step of the ith proposal in the current batch, and t_j denotes the training step when the jth proposal was added to the memory bank. w_0 is a hyperparameter specifying the base weight for the most recent proposal, and α is the decay factor.

D. Instance Masking Module

During the fine-tuning stage of the FSOD, the limited number of selected instances (shots) can lead to labeling errors. This issue arises when only some objects from one image are labeled, while all other objects are designated as background, regardless of whether they belong to base or novel classes. This challenge is depicted in the middle part of Fig. 4, where instances with incorrect labels confuse the few-shot detector.

To address this problem, we implement an instance masking module in the data preparation phase. In scenarios where an image contains multiple object instances, but only a subset is selected for few-shot training, we apply a Gaussian blurring operation to mask out all unselected objects if the object is not overlapped with any other selected objects, irrespective of their category, see the right part of Fig. 4. This technique ensures that these unselected instances are excluded from the few-shot fine-tuning process, thereby reducing potential confusion and improving detection performance.

E. Training Strategy

In the base-training phase, we train the model to detect objects from base classes. The overall loss consists of two parts, object classification loss ($\mathcal{L}_{\text{cls}}$), and OBB regression loss ($\mathcal{L}_{\text{reg}}$). After the base-training phase, we modified the classification subnet to predict class labels in $C_{\text{base}} \cup C_{\text{novel}}$ during the fine-tuning phase. During the fine-tuning stage, we freeze the parameters of the backbone network and train other modules using the following loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + \lambda \mathcal{L}_{\text{MCL}} \tag{6}$$

where λ is a hyperparameter used to balance the MCL loss with detection losses. Note that the MCL module is only used in the training stage and does not introduce additional computational costs during inference.

IV. Experiments and Results

In this section, we present a comprehensive evaluation of our FOMC on three remote sensing image datasets. First, we introduce the datasets and outline the implementation details, followed by a discussion on hyperparameter optimization. Then, we demonstrate the superior performance of FOMC in terms of FSOOD performance. Finally, we conduct extensive ablation studies to validate the effectiveness of each proposed component.

A. Datasets

1) DOTA [8]: The DOTA benchmark dataset contains 2806 aerial images with OBB annotations. The size of these images ranges from 800 × 800 to 40 000 × 40 000 pixels. It consists of 188 282 instances from 15 categories: plane (PL), baseball diamond (BD), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). In our experiments, 1411 images from

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

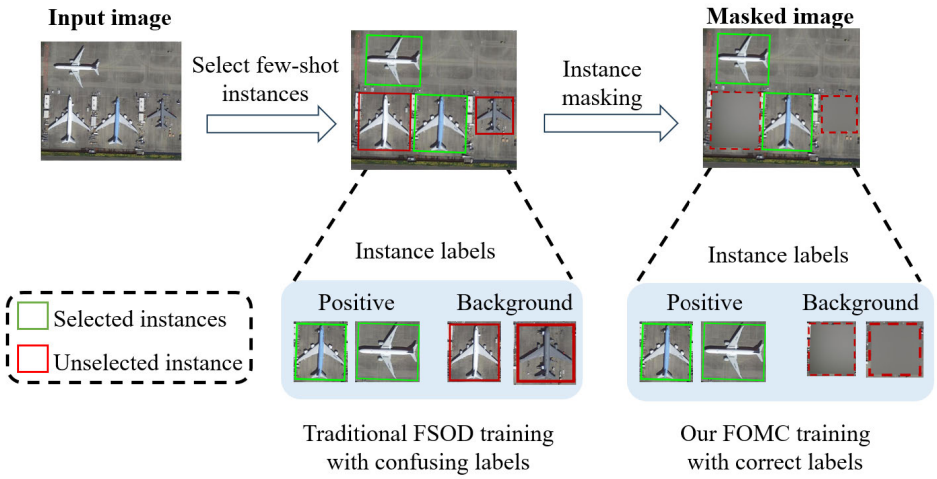


Fig. 4. Illustration of the instance masking module employed in our work. Previous FSOD methods treat unselected objects as background and therefore cause confusion for model training. Our instance masking module masks out all unselected objects using a Gaussian blurring operation.

the training set are used for model training, and 458 images from the validation set are used for evaluation. All training and validating images are cropped into 1024 × 1024 subimages before being fed into the network. For different split settings, the base classes include all classes except for the novel classes. For instance, in the split-1 setting, PL, LV, SH, GTF, and HA are chosen as novel classes, while the remaining 10 categories serve as base classes.

2) HRSC2016 [7]: HRSC2016 is designed for oriented ship detection with OBB annotations. It contains 1061 images and a total of 2976 ship instances. The size of these images ranges from 300 × 300 to 1500 × 900 pixels. As a fine-grained dataset, we focus on the most detailed L3 level categories. We exclude classes with fewer than 10 instances, resulting in a total of 20 classes. These classes are as follows: ship (SH), warcraft (WA), Nimitz (NI), enterprise (EN), Arleigh Burke (AB), Whidbey Island (WI), Perry (PE), San Antonio (SAN), Ticonderoga (TIC), container (CT), car carrier A (CCA), hovercraft (HC), container A (CTA), midway class (MC), submarine (SUB), Tarawa (TAR), Austen (AUS), commander A (CMA), medical (MED), and car carrier B (CCB).

3) DIOR-R [16]: DIOR-R is a new OBB detection dataset that shares the same images with DIOR [56] dataset. It contains 23 462 images and 192 518 instances, with 5862 images in the train set, 5863 images in the validation set, and 11 738 images in the test set. The image size is 800 × 800 pixels. In our experiments, we use the training set for model training and the validation set for evaluation. This dataset includes 20 common categories, including airplane (APL), airport (APO), baseball field (BF), basketball court (BC), bridge (BR), chimney (CH), expressway service area (ESA), expressway toll station (ETS), dam (DAM), golf field (GF), GTF, harbor (HA), overpass (OP), ship (SH), stadium (STA), storage tank (STO), TC, train station (TS), vehicle (VE), and windmill (WM).

B. Implementation Details

In our experiments, we choose both S^2A-Net and Oriented R-CNN as base detectors. Unless otherwise specified, S^2A-Net serves as the default base detector in our implementation. The base detectors adopt ResNet-50 [57] with a feature pyramid network (FPN) for the hierarchical feature extraction. The model is optimized using a standard stochastic gradient descent (SGD) optimizer with a momentum of 0.9 and a weight decay of $1e^{-4}$. For the base-training stage, the model is trained for 18 epochs on the DOTA dataset. The initial learning rate is set to 0.01 and decreased by a factor of 0.1 at epochs 11 and 16. During the fine-tuning stage, the initial learning rate is set to $1e^{-3}$ and decreased by a factor of 0.1 at 8000 iterations. For the HRSC2016 dataset, we train the base model for 36 epochs. The initial learning rate is set to 0.01 and decreases by a factor of 0.1 at epochs 33 and 24. During the fine-tuning stage, the initial learning rate is set to $1e^{-3}$. For the DIOR-R dataset, we train the base model for 12 epochs. The initial learning rate is set to 0.01 and decreases by a factor of 0.1 at epochs 8 and 11. During the fine-tuning stage, the initial learning rate is set to $1e^{-3}$.

A learning rate warm-up strategy is implemented for the first 200 iterations. The batch size is set to 8 during the base-training stage and 12 during the fine-tuning stage. The hyperparameters of the MCL module are set as follows: $\lambda = 0.5$, $w_0 = 0.95$, $\alpha = 0.01$, and the memory bank size $N = 8192$.

In both the base-training and fine-tuning stages, the input images are augmented by randomly applying horizontal flipping and rotation. For the evaluation metrics, we adopt the AP_{50} metric from the VOC2007 benchmark [58] to evaluate the performance of FSOOD. Predictions with an IoU greater than 0.5 with ground truth are considered true positives. All experiments are conducted using an NVIDIA GeForce RTX 3090Ti GPU.

C. Hyperparameter Selection

1) Loss Weight λ in the MCL Module: We first investigate how detection performance varies with different loss weights in the MCL module. Specifically, we conduct experiments with (λ) set to 0.05, 0.3, 0.5, and 0.8 on the DOTA dataset using a 10-shot setting. As shown in Table I, performance improves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

TABLE I
10-SHOT DETECTION PERFORMANCE ON THE DOTA DATASET WITH DIFFERENT CONFIGURATIONS OF MEMORY BACK SIZE (N) AND LOSS WEIGHT (λ)

N	λ	PL	LV	SH	GTF	HA	Avg.
-	-	11.4	20.9	24.7	12.5	0.4	14.0
-	0.05	11.2	38.0	21.4	13.5	1.8	17.2
-	0.3	**11.7**	38.8	21.1	13.4	1.6	17.3
-	0.5	11.5	37.4	25.5	12.9	**1.9**	**17.9**
-	0.8	**11.7**	38.3	24.6	12.8	1.3	17.7
2048	0.5	10.8	33.3	**30.4**	14.0	1.2	18.0
4096	0.5	**11.7**	37.3	27.8	13.4	1.5	18.4
8192	0.5	11.3	**38.9**	27.6	**14.1**	1.2	**18.8**
16384	0.5	**11.7**	37.7	27.4	13.9	1.4	18.4

as (λ) increases from 0.05, plateaus at $\lambda = 0.5$, and saturates when λ reaches 0.8. Based on these results, we set λ to 0.5 for the subsequent experiments.

2) Memory Bank Size: We further investigate the effect of varying memory bank sizes on our method's performance. Specifically, we train our model with different memory bank sizes N in the MCL module and evaluate its performance on the DOTA dataset using a 10-shot setting. In our experiments, we test memory bank sizes of 2048, 4096, 8192, and 16 384. As shown in the ablation results in Table I, detection accuracy improves modestly as the memory bank size increases, with $N = 8192$ providing the optimal performance. Although our MCL module benefits from larger memory banks, we opt not to use values of N greater than 8192 to avoid the additional computational overhead during training. In the following experiments, we set N to 8192 by default.

D. Comparing Methods

We mainly compare the FSOOD performance with the following methods in our experiments.

1) *S^2A-Net [59] + TFA [40]:* S^2A-Net is a prevalent one-stage method for oriented object detection, originally designed for the fully supervised object detection task. To adapt it for FSOOD, we follow the procedure of TFA [40], where we first train S^2A-Net on the base classes and then fine-tune it on the novel classes using a few-shot learning setup. This approach helps to leverage the knowledge learned from the base classes and transfer it to the novel classes while minimizing overfitting.
2) *Oriented R-CNN [36] + TFA [40]:* Oriented R-CNN is a prevalent two-stage detector and one of the state-of-the-art (SOTA) methods for oriented object detection. We adapt this method for few-shot learning by employing the same TFA strategy. This adaptation allows Oriented R-CNN to effectively generalize to novel classes with limited examples by using the feature reuse mechanism introduced by TFA, thus enhancing few-shot detection performance.
3) *R3Det [60] + TFA [40]:* R3Det is an advanced single-stage detector designed for oriented object detection, featuring a refined feature refinement mechanism to improve localization accuracy. We follow the TFA strategy to adapt it for FSOOD, where R3Det is first trained on the base classes and then fine-tuned on the novel classes.

E. FSOOD Results

1) DOTA: Table II presents the performance comparison on the novel classes of the DOTA dataset. Our method with an S^2A-Net base detector demonstrates better performance across different splits and shot settings. For example, in split-1, our method shows a 5.3% improvement over the baseline S^2A-Net model and outperforms the SOTA Oriented R-CNN by 3.4% in the 30-shot setting. We present the per-class detection performance in Table III. As shown, our method significantly outperforms the baseline S^2A-Net model in multiple categories, including plane, GTF, BD, BC, roundabout, and swimming pool. Specifically, for the baseball BD, our method achieves a 19.4% improvement over S^2A-Net and a 7.6% improvement over the Oriented R-CNN model. It is also worth noting that all methods perform poorly on the harbor class. This is likely due to the large diversity in aspect ratios of harbor objects, which presents a considerable challenge for the few-shot detection task. Additionally, per-class performance and COCO-style metrics are reported in the Supplementary Material.

We further investigate whether fine-tuning the model on novel classes affects its detection performance on base classes by examining Table IV. The results show that our FOMC method consistently outperforms comparing methods across all shot settings and achieves competitive results compared to Oriented R-CNN. Notably, there are two important findings as follows.

1) Our method with S^2A-Net base detector achieves better performance on novel classes than previous methods, as indicated by nAP scores.
2) Our method with Oriented R-CNN base detector achieves even better performance and is significantly better than TFA-based methods. Specifically, in the 30-shot setting, our method with Oriented R-CNN base detector attains an nAP score of 30.1%, while its TFA-based counterpart only gets an nAP score of 20.5.
3) Our method maintains strong performance on base categories (as measured by bAP) and overall detection performance (AP), confirming that FOMC effectively detects novel-class objects without substantially compromising base-class detection. These results collectively demonstrate that our FOMC method successfully addresses the FSOOD challenge while preserving base-class performance.

Furthermore, we provide visualizations of the 20-shot oriented object detection results of FOMC in Fig. 5. These results demonstrate that our FOMC model can accurately detect oriented objects with complex backgrounds, even with limited training data. This highlights the effectiveness of FOMC in capturing key features and improving detection performance in challenging few-shot scenarios.

2) HRSC2016: The quantitative few-shot detection performance on the HRSC2016 dataset is presented in Table V. Our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

TABLE II
FSOOD PERFORMANCE ON NOVEL CLASSES OF THE DOTA DATASET. THE † DENOTES THAT THE RESULTS ARE BORROWED FROM [49]

Method	split-1				split-2				split-3			
	5	10	20	30	5	10	20	30	5	10	20	30
RoI Transformer + TFA†	10.4	12.8	18.4	20.9	13.6	21.8	33.9	**40.6**	**10.6**	14.5	**21.7**	22.8
ReDet + TFA†	10.5	14.8	18.9	20.3	12.1	17.8	34.4	42.1	9.9	14.1	20.9	21.5
Oriented R-CNN + TFA†	12.1	14.2	18.1	20.4	14.2	22.8	33.6	39.6	10.4	15.3	21.4	22.6
R3Det + TFA	6.2	8.6	12.2	14.9	7.0	15.4	24.6	29.5	8.4	12.1	14.9	18.6
S^2A-Net + TFA	8.0	10.8	13.3	18.5	10.4	17.1	28.5	33.9	8.7	11.3	16.2	20.6
Ours w/ S^2A-Net	**12.8**	**17.0**	**20.3**	**23.8**	**16.6**	**25.3**	**35.8**	40.4	10.5	**16.6**	19.8	**24.0**

TABLE III
30-SHOT DETECTION PERFORMANCE FOR DIFFERENT FEW-SHOT SPLIT SETTINGS. THE † DENOTES THAT THE RESULTS ARE BORROWED FROM [49]

Method	split-1					split-2					split-3				
	PL	LV	SH	GTF	HA	ST	BD	BC	TC	RA	BR	SV	HC	SBF	SP
RoI Transformer + TFA†	33.4	19.8	9.1	**31.3**	10.6	40.7	39.6	22.4	57.6	42.7	12.1	23.2	**29.7**	29.4	19.9
ReDet + TFA†	**35.4**	17.1	9.1	29.3	**11.8**	34.2	31.3	**28.2**	**71.4**	**45.6**	**15.2**	19.8	27.7	29.9	15.4
Oriented R-CNN + TFA†	34.2	**24.0**	9.5	26.1	8.3	39.2	41.4	20.3	51.5	45.4	12.1	20.3	28.5	**31.3**	20.5
R3Det + TFA	22.9	11.0	9.1	21.3	10.4	21.1	38.1	10.0	51.2	26.9	9.2	11.2	17.4	27.3	28.2
S^2A-Net + TFA	22.9	19.7	**34.9**	15.0	0.2	43.8	29.6	9.3	61.8	24.9	9.2	**34.9**	1.5	23.4	34.0
Ours w/ S^2A-Net	35.1	18.9	32.9	27.5	4.4	**41.9**	**49.0**	18.1	54.3	38.9	9.5	34.6	3.7	22.5	**50.0**

TABLE IV
FEW-SHOT DETECTION RESULTS ON DOTA DATASET UNDER 5, 10, 20, 30-SHOT SETTINGS IN THE SPLIT-1 SETTING. BAP, NAP, AND AP REPRESENT MAP FOR BASE CLASSES, NOVEL CLASSES, AND ALL CLASSES, RESPECTIVELY. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Shot	Method	bAP	nAP	AP
5	Oriented R-CNN + TFA	47.4	11.2	35.3
	R3Det + TFA	37.9	6.2	27.3
	S^2A-Net + TFA	38.4	8.0	28.3
	Ours w/ S^2A-Net	43.6	12.8	33.3
	Ours w/ Oriented R-CNN	**57.4**	**19.1**	**44.6**
10	Oriented R-CNN + TFA	50.9	14.7	38.8
	R3Det + TFA	37.1	8.6	27.6
	S^2A-Net + TFA	35.5	10.8	27.2
	Ours w/ S^2A-Net	43.9	17.0	33.0
	Ours w/ Oriented R-CNN	**57.3**	**22.7**	**45.7**
20	Oriented R-CNN + TFA	46.0	17.7	36.6
	R3Det + TFA	37.5	12.2	29.1
	S^2A-Net + TFA	36.8	13.3	29.0
	Ours w/ S^2A-Net	46.3	20.3	37.6
	Ours w/ Oriented R-CNN	**62.6**	**25.9**	**50.4**
30	Oriented R-CNN + TFA	53.7	20.5	42.7
	R3Det + TFA	42.4	14.9	33.2
	S^2A-Net + TFA	42.5	18.5	34.5
	Ours w/ S^2A-Net	48.3	23.8	40.1
	Ours w/ Oriented R-CNN	**61.4**	**30.1**	**51.0**

FOMC method consistently outperforms competing methods across all shot settings and evaluation metrics. For example, in the 3-shot setting, our method with S^2A-Net and Oriented R-CNN base detector achieves nAP scores of 42.2% and 47.2%, which are significantly better than their TFA counterparts. As the number of shots increases, the detection performance on novel classes exhibits rapid improvement. Specifically, with S^2A-Net base detector, our method attains nAP scores of 42.4%, 55.1%, and 66.8% under the 3-shot, 5-shot, and 10-shot settings, respectively. Meanwhile, our method maintains strong performance on base categories, with bAP scores of 66.8% and 69.6% with S^2A-Net and Oriented R-CNN base detectors in the 3-shot setting, respectively. These results further validate that our method can enhance the performance of novel classes while maintaining strong detection capabilities in base classes. The oriented ship detection results under the 10-shot setting are visualized in Fig. 6. As shown in Fig. 6, our model can detect ships of different sizes, appearances, and aspect ratios, even with limited training data.

3) DIOR-R: Table VI presents the few-shot detection performance on the DIOR-R dataset. Our FOMC method consistently delivers superior performance across different shot settings. In the 5-shot setting, our method with S^2A-Net base detector improves nAP score from 17.1% of TFA counterpart to 30.2%. With the Oriented R-CNN base detector, our method achieves the best nAP of 30.7%, as well as the best performance on base categories, as indicated by the best bAP score of 61.1%. As the number of shots increases, our method demonstrates rapid performance improvement on novel classes. Specifically, with the Oriented R-CNN base detector, our method attains nAP scores of 30.7%, 39.3%, and 49.0% for the 5-shot, 10-shot, and 20-shot settings. These results highlight the effectiveness of our approach in improving novel-class detection while maintaining strong base-class performance.

V. DISCUSSION

In this section, we perform comprehensive ablation studies to validate the efficacy of our proposed MCL framework and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

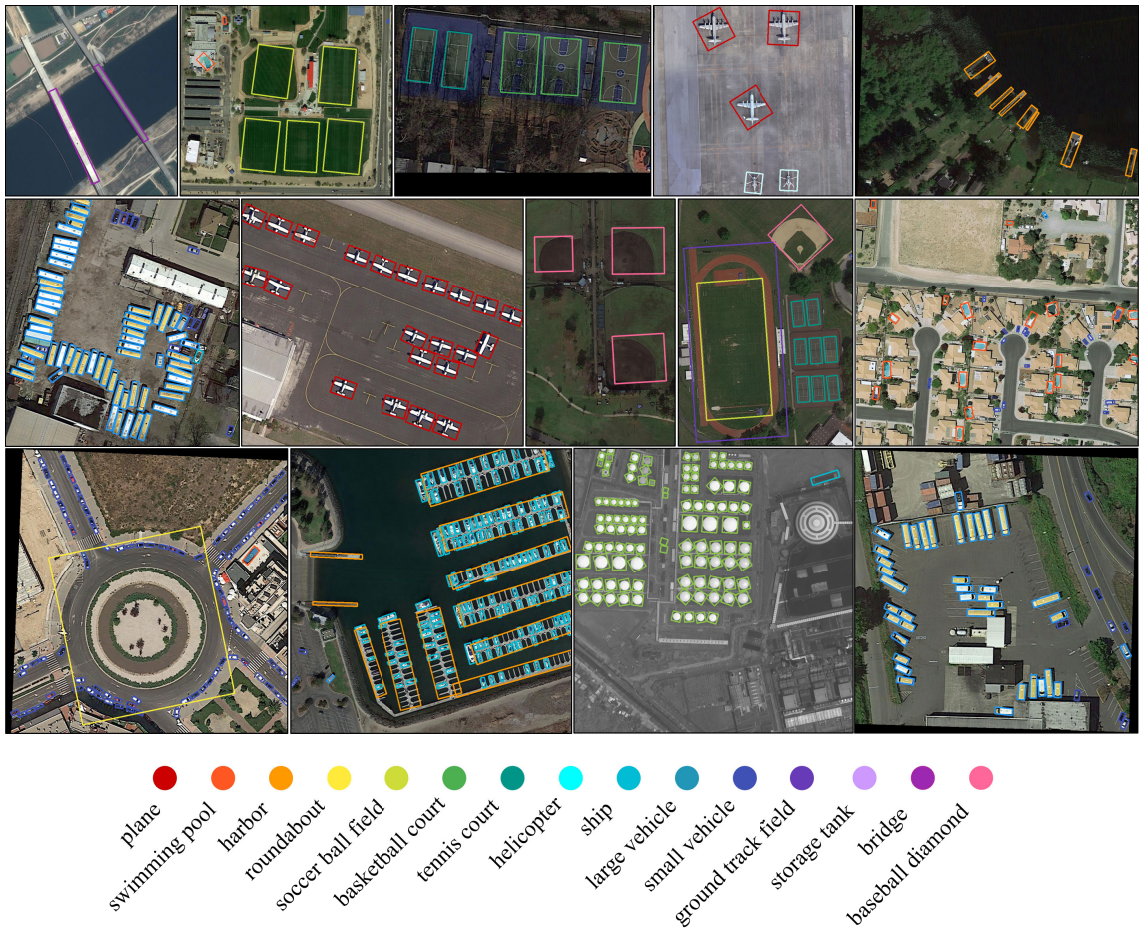


Fig. 5. Few-shot detection results on DOTA dataset. Our proposed model FOMC achieves satisfactory performance in detecting oriented objects in complex backgrounds, crowded and arbitrary-oriented instances for both novel and base classes.

TABLE V

FEW-SHOT DETECTION RESULTS ON HRSC2016 DATASET UNDER 3, 5, 10-SHOT SETTINGS. BAP, NAP, AND AP REPRESENT MAP FOR BASE CLASSES, NOVEL CLASSES, AND ALL CLASSES, RESPECTIVELY

Shot	Method	bAP	nAP	AP
3	Oriented R-CNN + TFA	67.0	34.5	58.9
	R3Det + TFA	46.4	17.0	39.1
	S^2A-Net + TFA	51.7	25.4	45.1
	Ours w/ S^2A-Net	66.8	42.4	60.7
	Ours w/ Oriented R-CNN	**69.6**	**47.2**	**64.0**
5	Oriented R-CNN + TFA	65.0	53.1	62.0
	R3Det + TFA	50.8	38.7	47.8
	S^2A-Net + TFA	56.9	48.2	54.7
	Ours w/ S^2A-Net	67.7	55.1	64.5
	Ours w/ Oriented R-CNN	**68.2**	**57.0**	**65.4**
10	Oriented R-CNN + TFA	68.5	63.0	67.2
	R3Det + TFA	58.6	44.1	55.0
	S^2A-Net + TFA	63.0	58.8	61.9
	Ours w/ S^2A-Net	67.0	**66.8**	66.9
	Ours w/ Oriented R-CNN	**69.4**	64.5	**68.1**

TABLE VI

FEW-SHOT DETECTION RESULTS ON DIOR-R DATASET UNDER 5, 10, 20-SHOT SETTINGS. BAP, NAP, AND AP REPRESENT MAP FOR BASE CLASSES, NOVEL CLASSES, AND ALL CLASSES, RESPECTIVELY

Shot	Method	bAP	nAP	AP
5	Oriented R-CNN + TFA	56.1	29.1	49.3
	R3Det + TFA	39.7	17.1	34.0
	S^2A-Net + TFA	41.0	18.4	35.3
	Ours w/ S^2A-Net	51.7	30.2	46.3
	Ours w/ Oriented R-CNN	**61.1**	**30.7**	**53.5**
10	Oriented R-CNN + TFA	49.1	34.6	45.5
	R3Det + TFA	42.8	28.1	39.1
	S^2A-Net + TFA	38.7	28.4	36.1
	Ours w/ S^2A-Net	59.2	37.9	53.9
	Ours w/ Oriented R-CNN	**61.7**	**39.3**	**56.1**
20	Oriented R-CNN + TFA	54.8	46.6	52.8
	R3Det + TFA	45.6	32.6	42.3
	S^2A-Net + TFA	42.9	33.8	40.6
	Ours w/ S^2A-Net	58.8	47.0	55.9
	Ours w/ Oriented R-CNN	**60.9**	**49.0**	**58.0**

instance masking module, followed by systematic investigations of model performance under varying instance selection strategies and few-shot configurations. Additionally, we conduct comparative analyses of computational efficiency and present representative failure cases to provide insights into the limitations of our approach.

A. Ablation Study

Table VII presents the results of the few-shot detection performance for the ablation of masking and MCL modules. The visualization results comparing S^2A-Net and FOMC are shown in Fig. 7. In the first column, S^2A-Net's few-shot

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

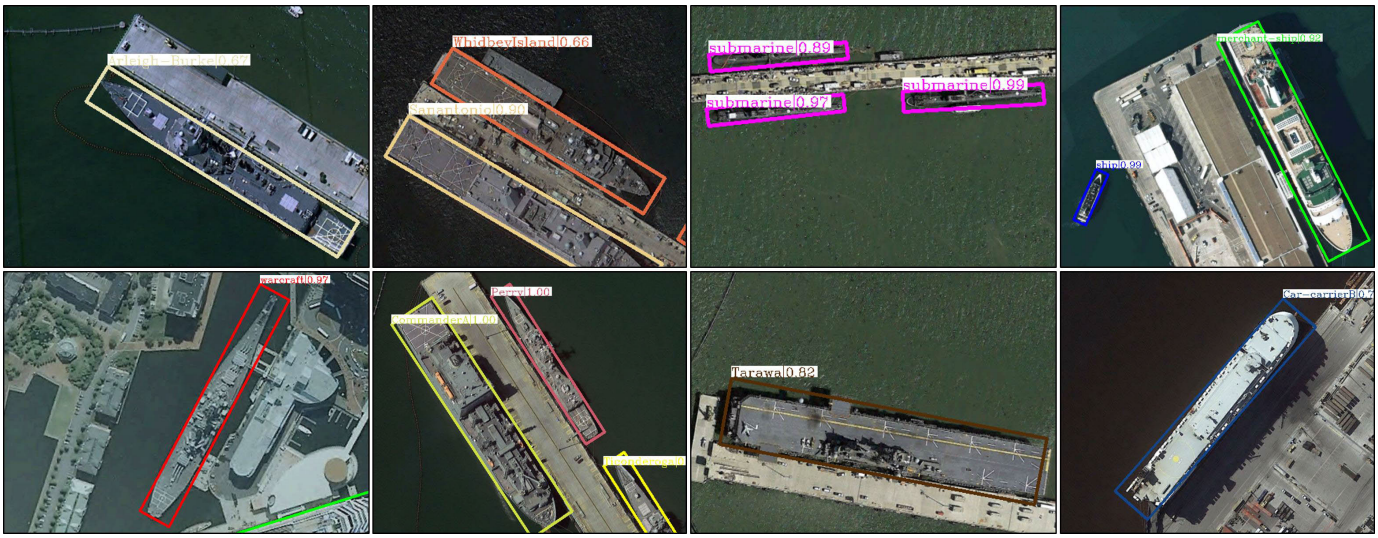


Fig. 6. Visualization of oriented ship detection results under 10-shot settings on the HRSC2016 dataset.

detection results exhibit many false positives in the harbor and BD class. It also fails to detect some ship instances in densely populated areas. In contrast, the second column shows that FOMC successfully detects these instances, with fewer false positives and missed detections compared to S^2A-Net.

1) Effect of MCL: By comparing the first and second rows of Table VII, the MCL module achieves a 21% relative improvement in the mAP of the novel classes compared to the baseline model. Our model achieves a notable performance improvement in three of the five novel classes, specifically LV, SH, and GTF. In particular, the incorporation of the MCL module results in a significant performance gain for the LV and GTF categories, while leading to a moderate performance decline in the airplane (PL) category. This decline is likely caused by misleading labels from unselected shots in airplane detection. Conversely, the improved performance on LV can be attributed to the MCL module's ability to better distinguish between visually similar classes, such as LVs and SVs. Overall, these findings highlight the effectiveness of the MCL module in enhancing the detection of novel object classes.

2) Effect of Instance Masking: We begin by examining the impact of our instance masking module on the baseline model. By comparing the first and third rows of Table VII, the introduction of the instance masking module leads to a notable improvement in detection performance, with the mAP of novel classes increasing from 18.5% to 22.5%. This performance boost demonstrates the effectiveness of the instance masking module in enabling the model to focus on essential object features while ignoring confusing unselected instances.

We also note that a similar masking strategy is proposed in SNIP [61] where a gradient masking strategy selectively backpropagates the gradients of object instances of different sizes as a function of the image scale. Based on this idea, we conduct experiments where gradients for specific object instances are selectively backpropagated based on the IoU between anchors and selected instances. The results are

TABLE VII

ABLATIVE PERFORMANCE OF THE MCL MODULE AND THE INSTANCE MASKING MODULE PROPOSED IN OUR METHOD. THE ^{g} INDICATES OUR INSTANCE MASKING IS REPLACED WITH GRADIENT MASKING [61]

Method	MCL	Masking	30-shot					
			PL	LV	SH	GTF	HA	Avg.
Baseline	×	×	22.9	19.7	34.9	15.0	0.2	18.5
Ours	✓	×	17.4	**41.2**	**35.2**	17.9	0.3	22.3
	×	✓	29.2	33.3	31.3	17.5	1.3	22.5
	✓	✓^{g}	30.3	29.9	32.0	19.2	**6.1**	23.5
	✓	✓	**35.3**	18.9	32.9	**27.5**	4.4	**23.8**

presented in the third row of Table VII. By comparing it with the baseline, we observe that the gradient masking strategy shows slightly better performance than our proposed instance masking strategy. However, it should be noted that the gradient masking strategy requires carefully setting the IoU threshold to select anchors for gradient calculations [61]. In contrast, our instance masking module discards objects that do not overlap with the selected support shots, eliminating the need for additional hyperparameters. It can be readily integrated into various few-shot detection methods without modifying the training objectives. However, in scenarios where ground objects exhibit strong spatial and contextual correlations—such as ships and harbors—this offline strategy may be less effective. In such cases, incorporating the online gradient masking mechanism could help mitigate performance degradation and improve results.

Moreover, by analyzing the bottom part of Table VII, we observe that our model benefits from the joint utilization of the proposed instance masking module and the MCL module. In this configuration, the average AP improves from 22.3% to 23.8%. This demonstrates that the instance masking module not only benefits the baseline model, but also synergizes with the MCL module to enhance the model's ability to learn more robust feature representations. This improvement is

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.



Fig. 7. Visualization of the oriented object detection results on the DOTA dataset under 20-shot setting with and without the MCL module. The baseline model struggles with classification in the few-shot setting. Our MCL module explicitly captures intraclass similarities and interclass differences, leading to more accurate object detection in few-shot scenarios compared to the baseline model.

particularly evident in the performance in challenging classes, e.g., harbor (HA), where the AP score increases from 0.3% to 4.4%, underscoring the robustness and generalizability of our proposed approach.

B. Robustness To Random Instance Selection

To demonstrate the robustness of our FOMC model against random instance selection, we repeat the experiments ten times, each time selecting instances from novel classes in a different random manner. These experiments are conducted on the DOTA dataset in a 30-shot setting. Our FOMC model consistently achieves an mAP of 21.6% $\pm$ 1.7% for the novel classes and 46.8% $\pm$ 2.3% for the base classes. These results highlight the robustness of our FOMC model under different instance selection scenarios. However, the random selection process does not explicitly control for instance diversity in terms of aspect ratios and orientations, which are particularly prominent in remote sensing images [49]. Investigating diversity-sensitive instance selection strategies could be a promising direction for future research.

C. Number of Shots

We evaluate the FSOOD performance of FOMC on novel classes across the DOTA, HRSC2016, and DIOR-R datasets with varying numbers of instances for fine-tuning. As shown in Fig. 8, the detection performance improves consistently as the number of shots increases. For the DOTA dataset, categories such as plane, LV, GTF, and ship exhibit a rapid increase in AP with more training instances. However, for the harbor class, the few-shot detection performance increases more slowly compared to other classes. This is likely due to its greater diversity of aspect ratios. For the HRSC2016 dataset, which focuses on oriented ship detection, FOMC achieves an accuracy comparable to the baseline model under fully supervised training with only 15-shot annotations, demonstrating its strong generalization capability with limited samples. Specifically, the few-shot detection performance of the MED class surpasses that of fully supervised training. Similarly, for the DIOR-R dataset, per-class AP trends indicate that novel-class detection performance steadily improves with an increasing number of shots, further validating the effectiveness of our approach across different domains and datasets.

TABLE VIII
COMPARISON OF COMPUTATIONAL EFFICIENCY METRICS WITH BATCH SIZE 1

Method	FLOPS (GB)	Params (MB)	Throughput (it/s)
R3Det	442.01	47.62	21.19
Ours w/ Oriented R-CNN	211.16	41.37	21.55
Ours w/ S^2A-Net	210.17	38.83	21.83

D. Model Efficiency

Table VIII compares the computational efficiency of different methods, including R3Det and our approach with two base detectors (Oriented R-CNN and S^2A-Net). From the table, our method demonstrates superior efficiency, with S^2A-Net achieving the lowest computational cost (210.17 GB FLOPS, 38.83 MB Params) and highest throughput (21.83 it/s), while R3Det is significantly more resource-intensive (442.01 GB FLOPS, 47.62 MB Params) and slower (21.19 it/s).

E. t-SNE Visualization

To understand learned feature representation, we visualize the per-class representations in Fig. 9. From this figure, the embedding space obtained by the baseline model struggles to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

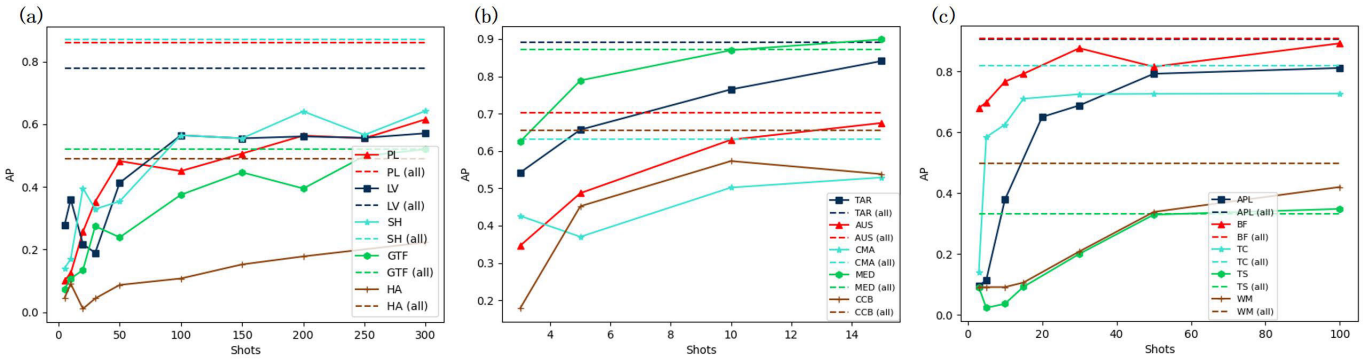


Fig. 8. Few-shot detection performance on novel classes with different numbers of shots. The results reported here are obtained using FOMC with the S^2A-Net detector. (a) DOTA dataset. (b) HRSC2016 dataset. (c) DIOR-R dataset. The horizontal dashed lines represent the performance of novel classes under fully supervised training.

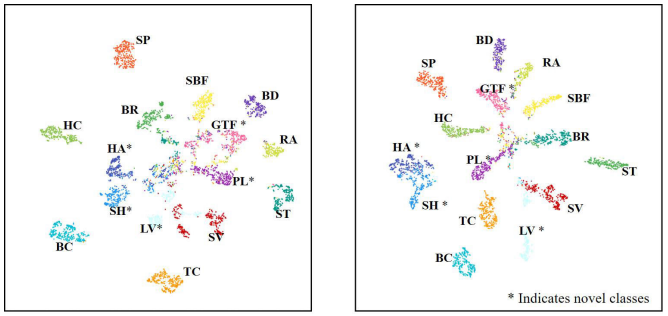


Fig. 9. t-SNE visualization of proposal embeddings obtained with and without our proposed MCL loss. The embedding space learned with the MCL loss exhibits a more compact intraclass feature distribution and a larger interclass margin. (a) Feature embeddings without MCL. (b) Feature embeddings will MCL.



Fig. 10. Failure cases of FOMC under the 30-shot setting (split-1) on the DOTA dataset.

distinguish proposal embeddings from different classes, due to the overlapped and entangled distribution of data points. Powered by our MCL module, feature embeddings from the same classes become more tightly clustered, with increased margins between different classes. This allows the model to better capture underlying semantics, leading to improved performance in the FSOOD task.

F. Failure Cases

Fig. 10 shows several failure cases from the 30-shot fine-tuning results of FOMC. In the top-left image, the bounding box for the GTF is misaligned with the ground truth. The top-right image illustrates missed detections for airplane instances, including a misclassification of an airplane as a helicopter. The bottom-left and bottom-right images further reveal missed detections and localization errors for the harbor and ship classes, respectively.

VI. Conclusion

In this study, we introduce the FOMC model, aiming to address the challenging oriented object detection task in few-shot settings. We design the MCL module, a contrastive learning-based approach with a memorable queue to learn more discriminative features, further improving the detection performance for overall classes. To mitigate the negative impact of confusing labels during the fine-tuning stage, we further propose the instance masking module to eliminate confusing labels from unselected instances. We conduct extensive experiments on three public benchmark datasets, DOTA, HRSC2016, and DIOR-R, to validate the effectiveness of our proposed methods.

Despite these advancements, few-shot learning for oriented object detection still faces challenges, particularly in reducing reliance on labeled data and improving generalization to unseen categories. Addressing these challenges presents opportunities for future research. The application of weakly supervised learning methods, such as PointOBB [62], which facilitates semisupervised learning from point annotations to OBB, could substantially reduce the dependence on labeled datasets. Combining few-shot learning models with these weakly supervised approaches represents a promising avenue for further exploration. Additionally, to enhance the effectiveness and robustness of visual feature extraction, leveraging vision and vision-language foundation models, such as masked autoencoders (MAEs) [63] and CLIP [64], could improve the performance of FSOD in remote sensing imagery.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[2] A. O. Ok and E. Baseski, "Circular oil tank detection from panchromatic satellite images: A new automated approach," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 6, pp. 1347–1351, Jun. 2015.

[3] X. Yu and Z. Shi, "Vehicle detection in remote sensing imagery based on salient information and local shape feature," *Optik*, vol. 126, no. 20, pp. 2485–2490, Oct. 2015.

[4] B. Sirmacek and C. Unsalan, "A probabilistic framework to detect buildings in aerial and satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 1, pp. 211–221, Jan. 2011.

[5] Y. Hu, X. Li, N. Zhou, L. Yang, L. Peng, and S. Xiao, "A sample update-based convolutional neural network framework for object detection in large-area remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 947–951, Jun. 2019.

[6] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2014, pp. 580–587.

[7] Z. Liu, L. Yuan, L. Weng, and Y. Yang, "A high resolution optical satellite image dataset for ship recognition and some new baselines," in *Proc. Int. Conf. Pattern Recognit. Appl. Methods*, vol. 2, 2017, pp. 324–331.

[8] G.-S. Xia et al., "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3974–3983.

[9] X. Sun, B. Wang, Z. Wang, H. Li, H. Li, and K. Fu, "Research progress on few-shot learning for remote sensing image interpretation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2387–2402, 2021.

[10] X. Li, J. Deng, and Y. Fang, "Few-shot object detection on remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021, Art. no. 5601614.

[11] S. Wolf, J. Meier, L. Sommer, and J. Beyerer, "Double head predictor based few-shot object detection for aerial imagery," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 721–731.

[12] G. Cheng et al., "Prototype-CNN for few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2021, Art. no. 5604610.

[13] Y. Zhou, H. Hu, J. Zhao, H. Zhu, R. Yao, and W.-L. Du, "Few-shot object detection via context-aware aggregation for remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[14] B. Sun, B. Li, S. Cai, Y. Yuan, and C. Zhang, "FSCE: Few-shot object detection via contrastive proposal encoding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7352–7362.

[15] Z. Liu, H. Wang, L. Weng, and Y. Yang, "Ship rotated bounding box space for ship extraction from high-resolution optical satellite images with complex backgrounds," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 8, pp. 1074–1078, Aug. 2016.

[16] G. Cheng et al., "Anchor-free oriented proposal generator for object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625411.

[17] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.

[18] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[19] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, Dec. 2015, pp. 91–99.

[20] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[21] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf.* Amsterdam, The Netherlands: Cham, Switzerland: Springer, Oct. 2016, pp. 21–37.

[22] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[23] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.

[24] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.

[25] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[26] X. Wu, D. Hong, and J. Chanussot, "UIU-net: U-net in U-net for infrared small object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 364–376, 2023.

[27] C. Li, B. Zhang, D. Hong, J. Yao, and J. Chanussot, "LRR-Net: An interpretable deep unfolding network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513412.

[28] L. Wen, Y. Cheng, Y. Fang, and X. Li, "A comprehensive survey of oriented object detection in remote sensing images," *Expert Syst. Appl.*, vol. 224, Aug. 2023, Art. no. 119960.

[29] J. Ding et al., "Object detection in aerial images: A large-scale benchmark and challenges," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7778–7796, Nov. 2022.

[30] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2849–2858.

[31] X. Yang et al., "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 8232–8241.

[32] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2020.

[33] W. Qian, X. Yang, S. Peng, X. Zhang, and J. Yan, "RSDet++: Point-based modulated loss for more accurate rotated object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 11, pp. 7869–7879, Nov. 2022.

[34] X. Yao, H. Shen, X. Feng, G. Cheng, and J. Han, "R^2IPoints: Pursuing rotation-insensitive point representation for aerial object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5623512.

[35] J. Han, J. Ding, N. Xue, and G. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2786–2795.

[36] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.

[37] X. Qian, B. Wu, G. Cheng, X. Yao, W. Wang, and J. Han, "Building a bridge of bounding box regression between oriented and horizontal object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605209.

[38] Y. Yao, G. Cheng, C. Lang, X. Xie, and J. Han, "Centric probability-based sample selection for oriented object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 4, pp. 3290–3302, Apr. 2025.

[39] Y.-X. Wang, D. Ramanan, and M. Hebert, "Meta-learning to detect rare objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9925–9934.

[40] X. Wang, T. E. Huang, T. Darrell, J. E. Gonzalez, and F. Yu, "Frustratingly simple few-shot object detection," 2020, *arXiv:2003.06957*.

[41] Z. Fan, Y. Ma, Z. Li, and J. Sun, "Generalized few-shot object detection without forgetting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4527–4536.

[42] Z. Xiao, J. Qi, W. Xue, and P. Zhong, "Few-shot object detection with self-adaptive attention network for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4854–4865, 2021.

[43] Z. Zhao, P. Tang, L. Zhao, and Z. Zhang, "Few-shot object detection of remote sensing images via two-stage fine-tuning," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[44] W.-Z. Li, J.-W. Zhou, X. Li, Y. Cao, and G. Jin, "Few-shot object detection on aerial imagery via deep metric learning and knowledge inheritance," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 122, Aug. 2023, Art. no. 103397.

[45] T. Zhang, X. Zhang, P. Zhu, X. Jia, X. Tang, and L. Jiao, "Generalized few-shot object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 353–364, Jan. 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.

[46] X. Lu et al., "Few-shot object detection in aerial imagery guided by text-modal knowledge," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604719.

[47] F. Zhang, Y. Shi, Z. Xiong, and X. X. Zhu, "Few-shot object detection in remote sensing: Lifting the curse of incompletely annotated novel objects," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603514.

[48] J. Wu, C. Lang, G. Cheng, X. Xie, and J. Han, "Retentive compensation and personality filtering for few-shot remote sensing object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 7, pp. 5805–5817, Jul. 2024.

[49] H. Su, Y. You, and S. Liu, "Multi-oriented enhancement branch and context-aware module for few-shot oriented object detection in remote sensing images," *Remote Sens.*, vol. 15, no. 14, p. 3544, Jul. 2023.

[50] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[51] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9729–9738.

[52] X. Chen, H. Fan, R. Girshick, and K. He, "Improved baselines with momentum contrastive learning," 2020, *arXiv:2003.04297*.

[53] P. Khosla et al., "Supervised contrastive learning," in *Proc. NIPS*, 2020, pp. 18661–18673.

[54] Y. Sun, Y. Chen, X. Wang, and X. Tang, "Deep learning face representation by joint identification-verification," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 27, Dec. 2014, pp. 1988–1996.

[55] L. Huang, S. Dai, and Z. He, "Few-shot object detection with dense-global feature interaction and dual-contrastive learning," *Int. J. Speech Technol.*, vol. 53, no. 11, pp. 14547–14564, Jun. 2023.

[56] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[57] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[58] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, 2010.

[59] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2021.

[60] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 3163–3171.

[61] B. Singh and L. S. Davis, "An analysis of scale invariance in object detection SNIP," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3578–3587.

[62] J. Luo, X. Yang, Y. Yu, Q. Li, J. Yan, and Y. Li, "PointOBB: Learning oriented object detection via single point supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, vol. 34, Jun. 2024, pp. 16730–16740.

[63] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.

[64] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 01:51:36 UTC from IEEE Xplore. Restrictions apply.